

- Place dots on the coil terminals in figure 1 to obtain correct dot conventions for the coupled coils.

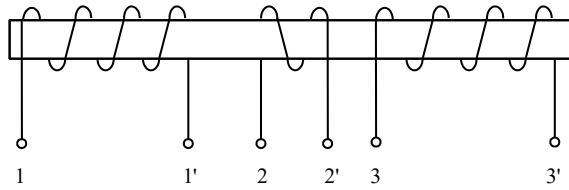


Figure 1

(Terminal 1, 2' and 3 are dot conventions.)

- Write loop equations in both time and phasor domain for the circuits in figure 2.

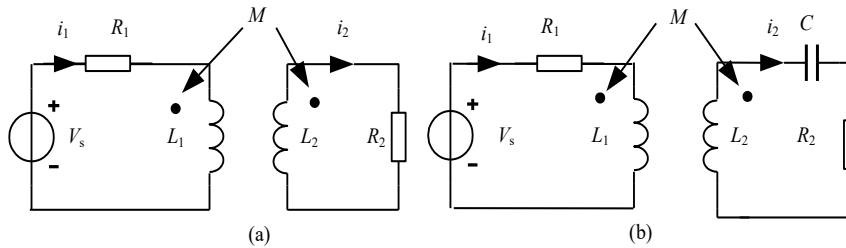


Figure 2

Solution

$$(a) \begin{cases} (R_1 + j\omega L_1) \dot{I}_1 - j\omega M \dot{I}_2 = \dot{V}_s \\ -j\omega M \dot{I}_1 + (R_2 + j\omega L_2) \dot{I}_2 = 0 \end{cases}$$

$$(b) \begin{cases} (R_1 + j\omega L_1) \dot{I}_1 - j\omega M \dot{I}_2 = \dot{V}_s \\ -j\omega M \dot{I}_1 + \left(R_2 + j\omega L_2 + \frac{1}{j\omega C} \right) \dot{I}_2 = 0 \end{cases}$$

- For the circuit of figure 3, determine the value of angular frequency ω such that the impedance $Z = r$. Take $L_1=0.01\text{H}$, $L_2=0.02\text{H}$, $C=2\mu\text{F}$ and $M=0.01\text{H}$.

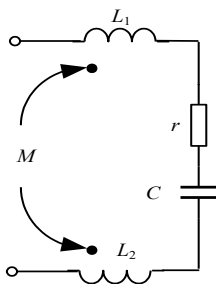


Figure 3

$$\omega = \frac{1}{\sqrt{(L_1 + L_2 - 2M)C}} = 0.707 \times 10^4 \text{ rad/s}$$

- Find input impedance Z_{ab} at angular frequency ω for each circuit in figure 4.

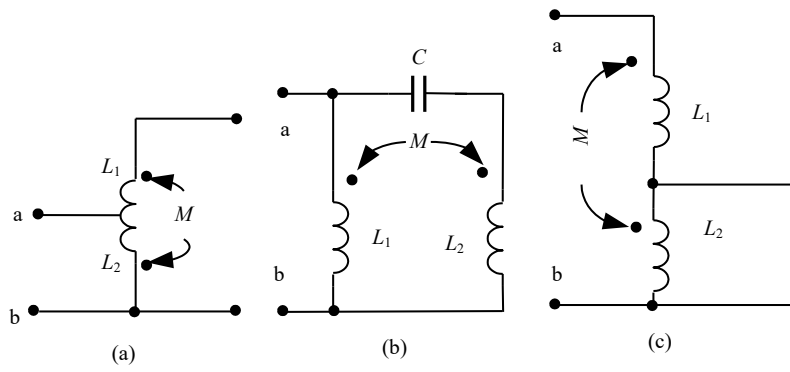


Figure 4

(a) $Z_i = j\omega L_2$

(b) $Z_i = j\omega M + j\omega(L_1 - M) // \left(j\omega(L_2 - M) + \frac{1}{j\omega C} \right)$

(c) $Z_i = j\omega \frac{L_1 L_2 - M^2}{L_2}$

5. Find $\dot{V}_2$ for the circuit in figure 5. Take $k=1/2$.

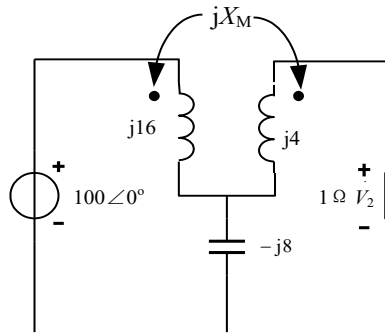


Figure 5

$\dot{V}_2 - \frac{50}{37}(1 + j6) = 8.22 \angle -99.5^\circ$

6. Find $\dot{I}_1$ in figure 6. Take $r = 10\Omega$, $L_1 = 4 \text{ mH}$, $L_2 = 6 \text{ mH}$, $M = 3 \text{ mH}$, $L_3 = 10 \text{ mH}$, $C = 100$

μF and $v_s = 100\sqrt{2} \cos(1000t + 60^\circ) (\text{V})$.

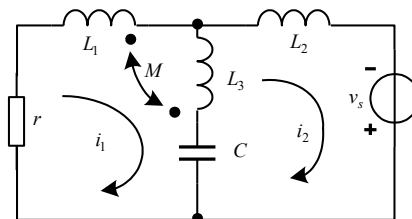


Figure 6

$$\dot{I}_1 = \frac{100\angle 60^\circ}{20 + j17} = 3.81\angle 19.6^\circ (\text{A})$$

7. Determine the equivalent inductance for the input impedance for the circuit in figure 7.

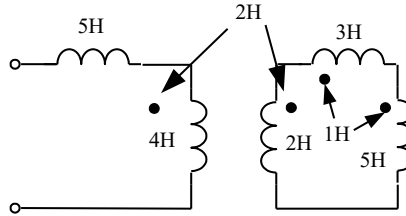


Figure 7

$$L_{eq} = 9 - \frac{1}{3} = 8.67 (\text{H})$$

8. Find the input impedance Z_i and the current phasor $\dot{I}_2$ in figure 8.

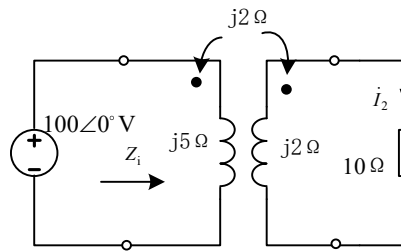


Figure 8

$$Z = \frac{5}{13} + j\frac{64}{13} = 4.9\angle 85.3^\circ (\Omega)$$

$$\dot{I}_2 = 4.01\angle -6^\circ (\text{A})$$

9. Find the power received by the resistor in secondary loop of the circuit in figure 9.

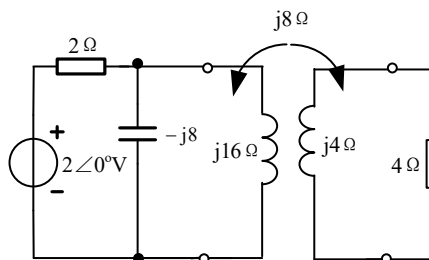


Figure 9

$$P_2 = 0.1562^2 \times 8 = 0.195 (\text{W})$$

10. Determine the Thevenin equivalent circuit of the circuit in Fig. 10 to the left of the terminals.

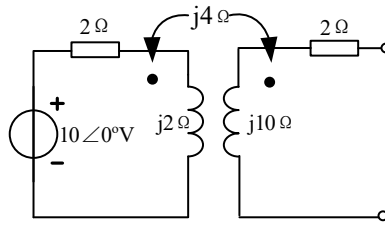


Figure 10

$$\dot{V}_{oc} = 10(1 + j) = 10\sqrt{2}\angle 45^\circ (\text{V})$$

$$Z_0 = 6 + j6(\Omega)$$

11. For the circuit in figure 11, determine the load impedance Z_L that draws maximum average power and find the power under the following restrictions to the load:

- (1) Z_L is a general impedance with resistive and reactive part.
- (2) Z_L is a resistor.

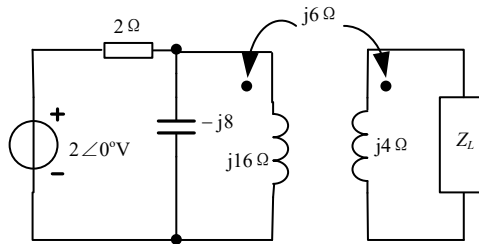


Figure 11

$$(1) Z_L = 0.277 - j1.72\Omega \quad P_{L_{\max}} = \frac{0.744^2}{4 \times 0.277} = 0.5\text{W}$$

$$(2) R_L = |Z_L| = \sqrt{0.277^2 + 1.72^2} \approx 1.74\Omega \quad P_{\max} \approx 0.14\text{W}$$

12. For the circuit in figure 12, determine the values of X_C and n such that the load Z receive maximum average power, and find this maximum power.

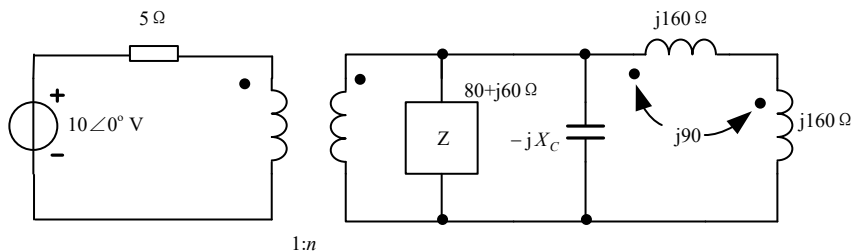


Figure 12

$$n = 5 \quad X_C = 125\Omega$$

$$P_{\max} = \frac{V_{\text{OC}}^2}{4R_0} = \frac{(10n)^2}{4 \times n^2 \times 5} = 5\text{W}$$

13. Calculate the average powers supplied by the source and absorbed by secondary load of the circuit in figure 13.

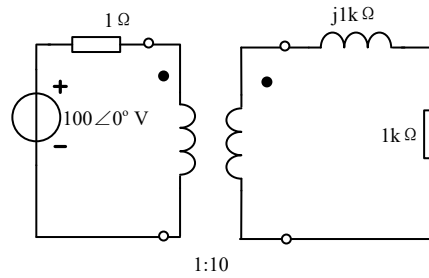


Figure 13

$$P_{V_s} = -498\text{W} \quad P_L = 453\text{W}$$

14. For the circuit in figure 14, find the turn ratio n of the matching transformer and the maximum average power drawn by the load R_L . Take $R_s = 10\text{k}\Omega$, $R_L = 3\Omega$ and $\dot{V}_s = 100\angle 0^\circ\text{V}$.

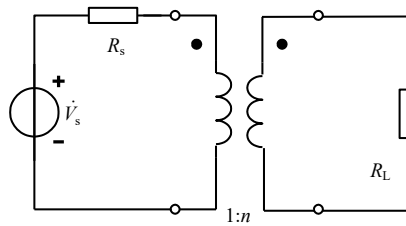


Figure 14

$$\therefore n = \frac{1}{58} \quad P_{L_{\max}} = 0.25\text{W}$$

15. Find the currents through all elements in the circuit of figure 15.

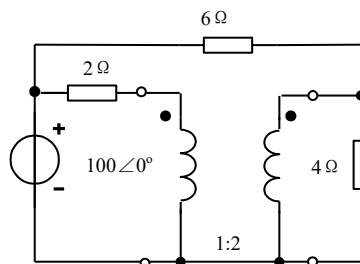


Figure 15

$$\dot{I}_{\text{source}} = 34.6\angle 0^\circ\text{A}, \quad \dot{I}_{\text{load}} = 19.2\angle 0^\circ\text{A}$$

$$P_{\text{load}} = 19.2^2 \times 4 = 1474.56(\text{W})$$

16. Determine the input impedance to the right of the terminals of the circuit in figure 16.

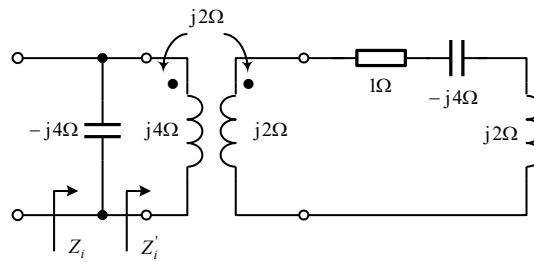


Figure 16

$$Z_i = 4 - j4 \Omega$$

17. Determine the turn ration n , the maximum average power drawn by R_L and the current of the secondary loop of the circuit in figure 17. Take $\dot{V}_s = 100\angle 0^\circ \text{ V}$, $\dot{I}_s = 100\angle 0^\circ \text{ A}$, $R_1 = R_2 = 1\Omega$ and $R_L = 10\Omega$.

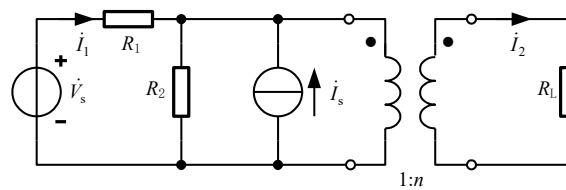


Figure 17

$$n = 2\sqrt{5}$$

$$P_{\max} = 5\text{kW}$$

$$I_2 = \frac{V_2}{R_L} = 10\sqrt{5}\angle 0^\circ \text{ V}$$

18. Find the input impedance for the circuit in figure 18. Take $n=0.5$.

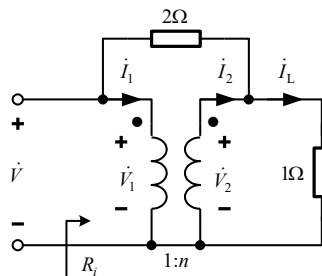


Figure 18

$$R_i = \frac{8}{3} \Omega$$

19. As shown in figure 19(a), the Thevenin equivalent of an audio signal source is connected to a load speaker of 8Ω .

- (1) Find the power drawn by the speaker, and show that this power is lower than the maximum power that a speaker could absorb from the source.
- (2) Insert a matching transformer between the source and the speaker to achieve maximum average power transfer to the speaker as shown in figure 19(b). Determine the turn ratio n , the input impedance seen from the source to the right hand side and the maximum power drawn by the speaker.

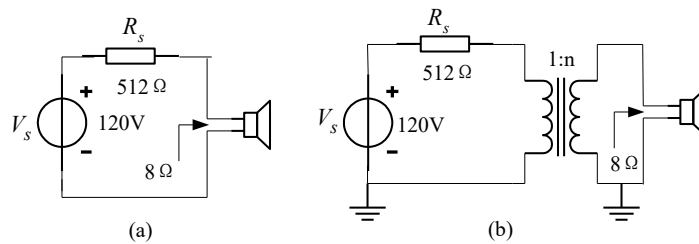


Figure 19

(1) $P = 0.43\text{W} < P_{\max}$

(2) $n = 1/8$